LECTURE-47 PROPERTIES OF LAPLACE TRANSFORM PART-01

Question-08

Find Laplace Transform & ROC for the following signal

(1) $x(t) = e^{-2t}\,u(t) + e^{-3t}\,u(t)$

(2) $x(t) = e^{-3t}\,u(t) + e^{2t}\,u(-t)$

(a) $e^{-2t}u(t) \longrightarrow \frac{1}{s+2}$ $\quad \sigma > -2$

$e^{-3t}u(t) \longrightarrow 1/s+3$ $\quad \sigma > -3$

$X(s) = \frac{1}{s+2} + \frac{1}{s+3}$; $\sigma > -2$

(b) $e^{-3t}u(t) \longrightarrow \frac{1}{s+3}$ $\quad \sigma > -3$

$e^{2t}u(-t) \longrightarrow \frac{-1}{s-2}$ $\quad \sigma < 2$

$X(s) = \frac{1}{s+3} - \frac{1}{s-2}$ $\quad -3 < \sigma < 2$

Question-09

Find Laplace Transform & ROC for the following signal

(1) $x(t) = e^{jat}\,u(t)$

(2) $x(t) = e^{2t}\,u(t) + e^{-3t}\,u(-t)$

(a) $e^{jat}u(t) \longrightarrow \dfrac{1}{s-ja}$ ROC: $\sigma > 0$

real part of pole

(b) $e^{2t}u(t) \longrightarrow \dfrac{1}{s-2}$ $\sigma > 2$

$e^{-3t}u(-t) \longrightarrow \dfrac{-1}{s+3}$ $\sigma < -3$

$\sigma > 2 \cap \sigma < -3$: null

LT does not exist

① $e^{jat}u(t) \longrightarrow \frac{1}{s-ja} \qquad \sigma>0$

$$= \frac{s+ja}{s^2+a^2} = \frac{s}{s^2+a^2} + j\frac{a}{s^2+a^2}$$

$$e^{jat}u(t) \longrightarrow \frac{s}{s^2+a^2} + j\frac{a}{s^2+a^2} \qquad \sigma>0$$

$$\cos at\, u(t) + j\sin at\, u(t) \longrightarrow \frac{s}{s^2+a^2} + j\frac{a}{s^2+a^2} \qquad \sigma>0$$

$$\cos at\, u(t) \longrightarrow \frac{s}{s^2+a^2} \qquad \sigma>0$$

$$\sin at\, u(t) \longrightarrow \frac{a}{s^2+a^2} \qquad \sigma>0$$

Question-10

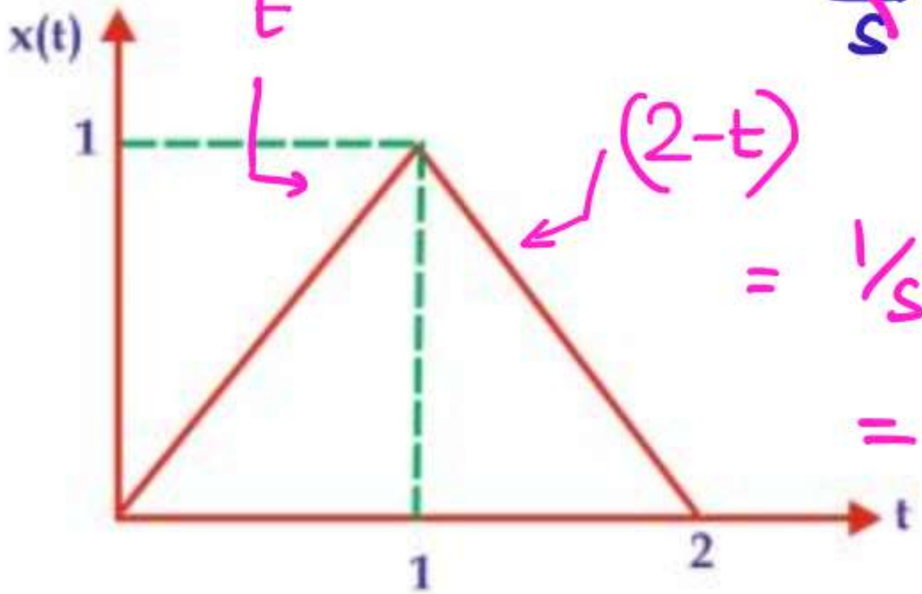
The Laplace Transform representation of the triangular pulse shown below is

(a) $\frac{1}{s^2}\left[1+e^{-2s}\right]$

(b) $\frac{1}{s^2}\left[1-e^{-2s}+e^{-2s}\right]$

(c) $\frac{1}{s^2}\left[1-e^{-s}+2e^{-2s}\right]$

(d) $\frac{1}{s^2}\left[1-2e^{-s}+e^{-2s}\right]$



$$X(s) = -\cancel{\frac{e^{-s}}{s}} - \frac{e^{-s}}{s^2} + \frac{1}{s^2} + \cancel{\frac{2e^{-s}}{s}} - \cancel{\frac{2e^{-2s}}{s}} + \cancel{\frac{2e^{-2s}}{s}} - \cancel{\frac{e^{-s}}{s}} + \frac{e^{-2s}}{s^2} - \frac{e^{-s}}{s^2}$$

$$= \frac{1}{s^2} - \frac{2e^{-s}}{s^2} + \frac{e^{-2s}}{s^2}$$

$$= \frac{1}{s^2}\left(1-2e^{-s}+e^{-2s}\right)$$

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}\,dt = \int_0^1 te^{-st}\,dt + \int_1^2 (2-t)e^{-st}\,dt$$

$$= \left[t\left[\frac{e^{-st}}{-s}\right] - \frac{e^{-st}}{s^2}\right]_0^1 + 2\left[\frac{e^{-st}}{-s}\right]_1^2 - \left[t\left[\frac{e^{-st}}{-s}\right]_1^2 - \frac{e^{-st}}{s^2}\right]_1^2$$

Question-11

Consider the following statements:

1. The Laplace transform of the unit impulse function is s × Laplace transform of the unit ramp function. x ramp: $1/s^2$ impulse = 1
2. The impulse function is a time derivative of the ramp function. x
3. The Laplace transform of the unit impulse function is s × Laplace transform of the unit step function. ✓ impulse = 1 step = $1/s$
4. The impulse function is a time derivative of the unit step function. ✓ Which of these statements is/are correct?

(a) 1 and 2 only

(c) 2 and 3 only

(b) 3 and 4 only ✓

(d) 1,2,3 and 4

Question-12

Find the Laplace transform of following signals and corresponding ROC.

(i) $x(t)=e^{-t}u(t)+e^{-2t}u(t)$

(ii) $x(t)=e^{-t}u(t)+e^{2t}u(-t)$

(iii) $x(t)=e^{t}u(t)+e^{-2t}u(-t)$

(a) $e^{-t}u(t) \rightarrow 1/s+1 \qquad \sigma > -1$

$e^{-2t}u(t) \rightarrow 1/s+2 \qquad \sigma > -2$

$X(s) = \frac{1}{s+1} + \frac{1}{s+2} \qquad \sigma > -1$

(b) $e^{-t}u(t) \rightarrow 1/s+1 \qquad \sigma > -1$

$e^{2t}u(-t) \rightarrow -1/s-2 \qquad \sigma < 2$

$X(s) = 1/s+1 - 1/s-2 \qquad -1 < \sigma < 2$

(c) $e^{t}u(t) \rightarrow \frac{1}{s-1} \qquad \sigma > 1$

$e^{-2t}u(-t) \rightarrow \frac{-1}{s+2} \qquad \sigma < -2$

LT does not exist

Some common Laplace Transform & ROC

**SN

Signal	Laplace Transform	ROC
$\delta(t)$	1	Entire s-plane
$u(t)$	$1/s$	Re {s} > 0
$t^n u(t)$	$n!/s^{n+1}$	Re{s} > 0
$e^{-at}\, u(t)$	$1/s+a$	Re{s} > -a
$e^{-at}\, u(-t)$	$-1/s+a$	Re{s} < -a
Cos at $u(t)$	$s/s^2 + a^2$	Re{s} > 0
Sin at $u(t)$	$a/s^2 + a^2$	Re{s} > 0
cosh at $u(t)$	$s/s^2 - a^2$	Re{s} > a
Sin h at $u(t)$	$a/s^2 - a^2$	Re{s} > a

Properties of Bilateral Laplace Transform

(1) Linearity

if $x_1(t) \longrightarrow X_1(s)$ ROC $= R_1$

$x_2(t) \longrightarrow X_2(s)$ ROC $= R_2$

then $a x_1(t) + b\, x_2(t) \longrightarrow a X_1(s) + b X_2(s)$ ROC: $R_1 \cap R_2$

if $R_1 \cap R_2 = \phi$ (null set)

then LT does not exist

if $a x_1(t) + b x_2(t)$: finite duration

then ROC is entire s-plane

except possible $\sigma = 0$ or $\sigma = \infty$

(2) Time shifting

if $x(t) \longrightarrow X(s)$ ROC = R

then $x(t-t_0) \longrightarrow e^{-st_0} X(s)$ ROC = R

proof

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

$$Y(s) = \int_{-\infty}^{\infty} x(t-t_0)\, e^{-st}\, dt$$

$t - t_0 = m$; $dt = dm$

$$Y(s) = \int_{-\infty}^{\infty} x(m)\, e^{-s(m+t_0)}\, dm$$

$$Y(s) = \int_{-\infty}^{\infty} x(m)\, e^{-st_0} e^{-sm}\, dm$$

$$= e^{-st_0} \int_{-\infty}^{\infty} x(m)\, e^{-sm}\, dm$$

$$= e^{-st_0} X(s)$$

Question-01

Determine Laplace Transform of $x(t) = u(t) - u(t-T)$?

$u(t) \longrightarrow 1/s$

$u(t-T) \rightarrow e^{-sT} \times 1/s$

$X(s) = \left[\frac{1-e^{-sT}}{s}\right]$

$\lim_{s\to 0} X(s) = \lim_{s\to 0} \left(\frac{1-e^{-sT}}{s}\right) = \frac{0}{0}$

$= \lim_{s\to 0} \frac{-Te^{-sT}}{1} = -T$

$s=0$ not a pole

$\lim_{s\to\infty} X(s) = 0$ $\quad s=\infty$ not a pole

finite duration signal $0 \to T$

ROC: entire s-plane

except possibly $s=0$ or $s=\infty$

$\lim_{s\to -\infty} X(s) = \infty$

ROC entire s plane

except $s=-\infty$

Question-02

Compute Bilateral Laplace Transform of following signals

(a) $x(t) = e^{2|t|}$

(b) $x(t) = e^{-2|t|}$

(a) $x(t) = e^{2|t|} = \begin{cases} e^{2t}, & t>0 \\ e^{-2t}, & t<0 \end{cases}$

$= e^{2t}u(t) + e^{-2t}u(-t)$

$e^{2t}u(t) \downarrow \frac{1}{s-2}$, $\sigma > 2$

$e^{-2t}u(-t) \downarrow -\frac{1}{s+2}$, $\sigma < -2$

$\sigma > 2 \cap \sigma < -2 = \phi$ LT does not exist

(b) $x(t) = \begin{cases} e^{-2t}, & t>0 \\ e^{2t}, & t<0 \end{cases}$

$x(t) = e^{-2t}u(t) + e^{2t}u(-t)$

$= \frac{1}{s+2} - \frac{1}{s-2}$

$\sigma > -2$ $\quad$ $\sigma < 2$

$= \frac{-4}{s^2-4}$ $\quad$ $-2 < \sigma < 2$

Question-03

Find Laplace Transform & ROC of following signals

(a) $x(t) = e^{-(t-3)}\, u(t-3)$

(b) $x(t) = e^{-2t}\{u(t) - u(t-5)\}$

(a) $e^{-t}u(t) \longrightarrow \dfrac{1}{s+1} \qquad \sigma > -1$

$e^{-(t-3)}u(t-3) \longrightarrow \dfrac{e^{-3s}}{s+1} \qquad \sigma > -1$

(b) $x(t) = e^{-2t}u(t) - e^{-2t}u(t-5)$

$= e^{-2t}u(t) - e^{-2(t-5)} \cdot e^{-10}u(t-5)$

$X(s) = \dfrac{1}{s+2} - e^{-10} \cdot \dfrac{e^{-5s}}{s+2}$

$= \dfrac{1}{(s+2)}\left[1 - e^{-5(s+2)}\right] \rightarrow s=-2$ is not a pole

ROC: entire s plane except $s = -\infty$

(3) Shifting in s-domain

if $x(t) \xrightarrow{LT} X(s)$, ROC = R

then $e^{s_0 t} x(t) \xrightarrow{LT} X(s-s_0)$; ROC shifts right by $Re(s_0)$

Proof

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

$$Y(s) = \int_{-\infty}^{\infty} e^{s_0 t} x(t)\, e^{-st}\, dt$$

$$= \int_{-\infty}^{\infty} x(t)\, e^{-(s-s_0)t}\, dt$$

$$= X(s-s_0)$$

Question-04

Determine Laplace Transform & ROC of following signals

(a) $x(t) = e^{-at} \cos bt\, u(t)$

(b) $x(t) = e^{-at} \sin bt\, u(t)$

$\cos bt\, u(t) \rightarrow \frac{s}{s^2+b^2} \quad \sigma > 0$

$\sin bt\, u(t) \rightarrow \frac{b}{s^2+b^2} \quad \sigma > 0$

(a) $e^{-at} \cos bt\, u(t) \longrightarrow \dfrac{(s+a)}{(s+a)^2+b^2}$ ROC: $\sigma + a > 0$

$\sigma > -a$

(b) $e^{-at} \sin bt\, u(t) \longrightarrow \dfrac{b}{(s+a)^2+b^2}$ ROC: $\sigma + a > 0$

$\sigma > -a$

(4) Time Reversal

if $x(t) \longrightarrow X(s)$, ROC: R

then $x(-t) \longrightarrow X(-s)$, ROC: replace σ by $-\sigma$

proof

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

$$Y(s) = \int_{-\infty}^{\infty} x(-t)\, e^{-st}\, dt$$

assume $-t = P$

$dt = -dp$

$$Y(s) = \int_{\infty}^{-\infty} x(p)\, e^{sP}\, (-dp)$$

$$Y(s) = \int_{-\infty}^{\infty} x(p)\, e^{sP}\, dp$$

$$= X(-s)$$

eg: $e^{-2t}u(t) \rightarrow \dfrac{1}{s+2}$ $\quad \sigma > -2$

$e^{2t}u(-t) \rightarrow \dfrac{1}{-s+2}$ $\quad -\sigma > -2$

$\sigma < 2$

(5) Time scaling

if $x(t) \longrightarrow X(s)$; ROC: R

then $x(at) \longrightarrow \frac{1}{|a|} X(s/a)$; ROC: replace σ by σ/a

proof

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

$$Y(s) = \int_{-\infty}^{\infty} x(at)\, e^{-st}\, dt$$

$at = \tau \qquad a\,dt = d\tau$

$$Y(s) = \int_{-\infty}^{\infty} x(\tau)\, e^{-s(\tau/a)}\, d\tau/a$$

$$Y(s) = \frac{1}{|a|} \int_{-\infty}^{\infty} x(\tau)\, e^{-(s/a)\tau}\, d\tau$$

$$= \frac{1}{|a|} X(s/a)$$

Question-05

Determine Bilateral Laplace Transform & ROC of following signals

(a) $x(t) = e^{t/3}\, u(t)$

$u(at) = u(t)$

$e^{t} u(t) \longrightarrow \frac{1}{s-1} \qquad \sigma > 1$

$e^{t/3} u(t/3) \longrightarrow \frac{1}{|1/3|} \times \frac{1}{\frac{s}{1/3} - 1} \quad ; \quad \frac{\sigma}{1/3} > 1$

$e^{t/3} u(t/3) \longrightarrow \frac{3}{3s-1} \qquad 3\sigma > 1$

$\sigma > 1/3$

$e^{t/3} u\left(t/3\right) = e^{t/3} u(t)$

Question-06

Consider a signal $x(t) = e^{-2t} u(t)$ & its Laplace Transform pair $x(t) \rightarrow X(s)$

(a) Inverse Laplace Transform of $X(S/2)$ is

(b) Inverse Laplace transform of $\{x(s-2) + x(s+2)\}$ is

① $x(at) \longrightarrow \frac{1}{|a|} X\left(\frac{s}{a}\right)$

$|a| x(at) \longrightarrow X\left(\frac{s}{a}\right)$

$2x(2t) \longrightarrow X\left(\frac{s}{2}\right)$

$2e^{-4t} u(2t) = 2e^{-4t} u(t)$

② $X(s-2) \rightarrow e^{2t} x(t)$

$X(s+2) \rightarrow e^{-2t} x(t)$

$X(s-2) + X(s+2) \rightarrow \left(e^{2t} x(t) + e^{-2t} x(t)\right)$

$= 2\cosh 2t \, x(t)$

$= 2\cosh 2t \cdot e^{-2t} u(t)$

or $\left(1 + e^{-4t}\right) u(t)$

(6) Differentiation in s-domain

if $x(t) \longrightarrow X(s)$ ROC = R

then $t x(t) \longrightarrow -\frac{d}{ds} X(s)$, ROC = R

$t^n x(t) \longrightarrow (-1)^n \frac{d^n X(s)}{ds^n}$ ROC = R

$t x(t) \longrightarrow -\frac{d}{d(j\omega)} X(j\omega)$

$-\frac{d}{j\,d\omega} X(j\omega)$

$j \frac{d}{d\omega} X(j\omega)$

Proof

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

$$\frac{d}{ds} X(s) = \int_{-\infty}^{\infty} x(t) \frac{d}{ds}(e^{-st}) dt$$

$$= \int_{-\infty}^{\infty} -t x(t) e^{-st} dt$$

$$-\frac{d}{ds} X(s) = \int_{-\infty}^{\infty} t x(t) e^{-st} dt$$

Question-07

If $x(t) = \dfrac{1}{s^2+s+1}$, Re{s} > -1/2 then if y(t) = tx(t) . Find Y(s)

$$tx(t) \longrightarrow -\frac{d}{ds}X(s) \qquad , \quad \sigma > -1/2$$

$$-\frac{d}{ds}\left(\frac{1}{s^2+s+1}\right)$$

$$\frac{1}{(s^2+s+1)^2} \times (2s+1)$$

Question-08

Find Laplace transform & ROC of following signals

(1) $x(t) = t\,u(t)$

(2) $x(t) = t^n u(t)$

$\hookrightarrow \frac{n!}{s^{n+1}}$

$u(t) \longrightarrow 1/s \qquad \sigma > 0$

$t u(t) \longrightarrow -\frac{d}{ds}\left(1/s\right) \qquad \sigma > 0$

$1/s^2 \qquad \sigma > 0$

$t^2 u(t) \rightarrow -d/ds\left(1/s^2\right) = 2/s^3 = 2!/s^3$

$t^3 u(t) \longrightarrow -d/ds\left(2/s^3\right) = 6/s^4 = 3!/s^4$

Question-09

Find Laplace transform & ROC of following signals

(a) $x(t) = te^{-at}\,u(t)$

(b) $x(t) = t^{n}e^{-at}\,u(t)$

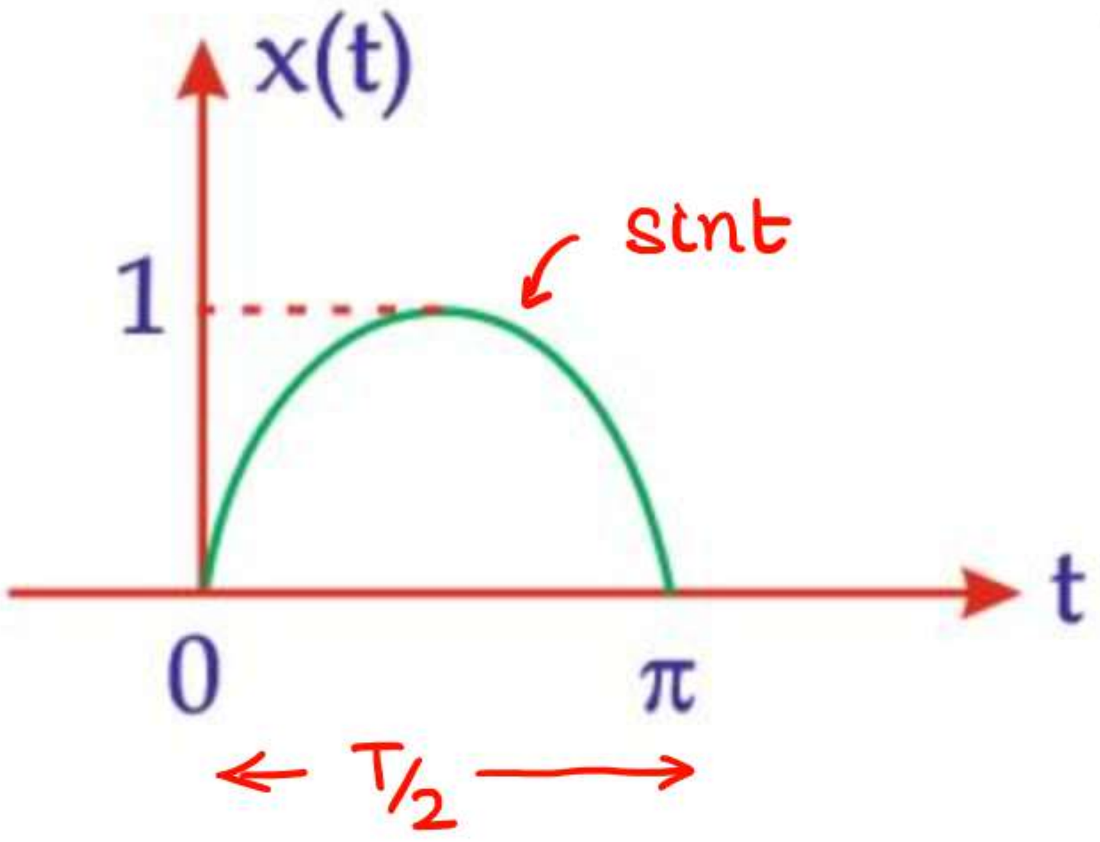
$\hookrightarrow \dfrac{n!}{(s+a)^{n+1}}$

(i) $e^{-at}u(t) \longrightarrow \dfrac{1}{(s+a)}$

$te^{-at}u(t) \longrightarrow -\dfrac{d}{ds}\left(\dfrac{1}{s+a}\right) = \dfrac{1}{(s+a)^2}$

Question-10

Find Laplace transform & ROC of following signal



$$T/2 = \pi$$

$$T = 2\pi$$

$$\omega_0 = 2\pi/T = 1$$

$$x(t) = \sin t \cdot [u(t) - u(t-\pi)]$$

$$= \sin t \cdot u(t) - \sin t \cdot u(t-\pi)$$

$$= \sin t \; u(t) + \sin(t-\pi)\, u(t-\pi)$$

$$X(s) = \frac{1}{s^2+1} + \frac{e^{-\pi s}}{s^2+1}$$

$$= \frac{(1+e^{-\pi s})}{s^2+1}$$

⑦ Convolution

if $x_1(t) \longrightarrow X_1(s)$, $ROC = R_1$

& $x_2(t) \longrightarrow X_2(s)$, $ROC = R_2$

then $x_1(t) * x_2(t) \longrightarrow X_1(s) \cdot X_2(s)$ $ROC = R_1 \cap R_2$

⑧ Multiplication

if $x_1(t) \longrightarrow X_1(s)$, $ROC = R_1$

& $x_2(t) \longrightarrow X_2(s)$, $ROC = R_2$

then $x_1(t) \cdot x_2(t) \longrightarrow \frac{1}{2\pi j}\left[X_1(s) * X_2(s)\right]$, $ROC = R_1 \cap R_2$

⑨ Integration in s-domain

if $x(t) \longrightarrow X(s) \quad ROC = R$

then $\dfrac{x(t)}{t} \longrightarrow \int_s^{\infty} X(\lambda)\,d\lambda \quad ROC = R$

proof

$$X(s) = \int_{-\infty}^{\infty} x(t)\,e^{-st}\,dt$$

$$\int_s^{\infty} X(\lambda)\,d\lambda = \int_s^{\infty}\int_{-\infty}^{\infty} x(t)\,e^{-\lambda t}\,dt\,d\lambda$$

$$= \int_{-\infty}^{\infty} x(t)\int_s^{\infty} e^{-\lambda t}\,d\lambda\,dt$$

$$\Rightarrow \int_{-\infty}^{\infty} x(t)\left[\frac{e^{-\lambda t}}{-t}\right]_s^{\infty} dt$$

$$= \int_{-\infty}^{\infty} \frac{x(t)}{t}\,e^{-st}\,dt$$

eg Find convolution of $x_1(t) = e^{-t}u(t)$ & $x_2(t) = u(t)$?

$X_1(s) = 1/(s+1)$ $\qquad X_2(s) = 1/s$

$$x_1(t) * x_2(t) \longrightarrow \frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1}$$

$$x(t) = (1 - e^{-t})\,u(t)$$

Question-11 **

Find inverse Laplace transform of following function

$$x(s) = \log\left(\frac{s+5}{s+6}\right)$$

$$\frac{d}{ds}X(s) = \left(\frac{s+6}{s+5}\right) \times \left[\frac{1(s+6) - 1(s+5)}{(s+6)^2}\right]$$

$$= \frac{1}{(s+5)(s+6)} = \frac{1}{(s+5)} - \frac{1}{(s+6)}$$

$$t x(t) \longrightarrow -\frac{d}{ds}X(s)$$

$$t x(t) = -\left[e^{-5t} - e^{-6t}\right]u(t)$$

$$x(t) = \frac{e^{-6t} - e^{-5t}}{t}\, u(t)$$

Question-12

Let $f(t) = x(t) + \alpha x(-t)$ where $x(t) = \beta e^{-t} u(t)$. $F(s) = \dfrac{s}{s^2 - 1}$

$-1 < \text{Re}\{s\} < 1$. Determine α & β

$X(s) = \beta / s+1$

$2\beta = 1 \quad \beta = 1/2$

$\alpha = -1$

$F(s) = X(s) + \alpha X(-s)$

$$\frac{s}{s^2-1} = \frac{\beta}{s+1} + \frac{\alpha\beta}{1-s} = \frac{\beta - s\beta + \alpha\beta s + \alpha\beta}{(1-s^2)}$$

$$= \frac{s(\beta - \alpha\beta) - (\alpha\beta + \beta)}{s^2-1}$$

$\beta - \alpha\beta = 1 \qquad \beta + \alpha\beta = 0$

Question-13

Determine Laplace transform of following signals

(a) $x(t) = \frac{\sin t}{t} u(t)$

(b) $x(t) = \left\{ \frac{e^{-at} - e^{-bt}}{t} \right\} u(t)$

HW

$\sin t \, u(t) \rightarrow \frac{1}{s^2+1}$

$\frac{\sin t}{t} u(t) \longrightarrow \int_s^{\infty} \frac{1}{\lambda^2+1} d\lambda$

$= [\tan^{-1} \lambda]_s^{\infty}$

$= \tan^{-1}\infty - \tan^{-1} s$

$= \pi/2 - \tan^{-1} s$

$= \cot^{-1} s$

Question-14

If $x(t) = 2\sqrt{\frac{t}{\pi}} \rightarrow s^{-3/2}$. Then find Laplace transform of $\sqrt{\frac{1}{\pi t}}$?

HW